

Light Project Spec:

General description of the project

This project is an interactive Kansas City event discovery application designed to help residents, students, and tourists find things to do in one place. Rather than searching multiple websites, users can browse concerts, festivals, museums, restaurants, coffee shops, parks, and other local activities through a single platform. The goal is to simplify event discovery and make it easier for people to explore the Kansas City area based on their interests, budget, and location.

The application will collect data from multiple sources, including the Ticketmaster Discovery API, Eventbrite, OpenStreetMap, and other publicly available datasets. Information such as event names, dates, locations, categories, and prices will be processed and combined into a searchable database. Users will be able to filter activities by date, cost, category, or distance, while the system removes duplicate events and organizes information into a user-friendly format.

A key feature of the project is an interactive map built with Leaflet and OpenStreetMap. Instead of only displaying individual event pins, the map will help users explore Kansas City neighborhoods such as Crossroads, River Market, Westport, and the Plaza. Events, attractions, and activity hotspots will be visualized through map markers, neighborhood overlays, and optional heatmaps, allowing users to quickly identify areas that match their interests and discover new experiences throughout the city.

Task Vignettes (User activity "flow")

Vignette 1: Discovering Events Through the Interactive Map

A college student is looking for something to do on a Friday night but does not know where to start. They open the Kansas City Event Explorer and are presented with an interactive map showing neighborhoods such as Crossroads, River Market, Westport, and the Plaza. The student selects the "Live Music" category and immediately sees upcoming concerts, local performances, and music festivals displayed on the map. As they zoom in, event clusters separate into individual markers, making it easier to identify activities near their preferred area.

The student clicks on an event marker in the Crossroads district and views additional details, including the event description, date, time, location, and ticket information. After comparing a few options, they save an event to their favorites list for later review. The application successfully reduces the effort required to search multiple websites and helps the user quickly identify entertainment opportunities nearby.

Technical Details

- Retrieve event data from Ticketmaster and Eventbrite APIs.
- Store event names, categories, locations, dates, and ticket links.
- Use Leaflet.js and OpenStreetMap to display an interactive map.

- Convert event locations into latitude and longitude coordinates.
- Cluster nearby markers to reduce visual clutter.
- Save favorited events in a database or browser storage.

Vignette 2: Exploring a Neighborhood

A tourist visiting Kansas City for the weekend wants to find a neighborhood with multiple activities within walking distance. Instead of searching event-by-event, they select the River Market neighborhood on the map. The application highlights local attractions, restaurants, coffee shops, and upcoming events occurring within that district. The user learns that a farmers market, a local festival, and several highly rated restaurants are all located nearby.

The user reviews the neighborhood overview and decides that River Market offers enough activities for an entire afternoon. They use the map to estimate walking distances between attractions and save several locations to their itinerary. By organizing information around neighborhoods rather than individual events, the application helps users understand the city and plan their visits more effectively.

Technical Details

- Import neighborhood boundary data using GeoJSON files.
- Display neighborhood overlays on the map.
- Associate events and attractions with neighborhood polygons.
- Pull restaurant and attraction data from OpenStreetMap datasets.
- Calculate distances between points of interest.
- Provide neighborhood summary information when selected.

Vignette 3: Finding Personalized Recommendations

A local resident wants affordable activities for the weekend. Upon opening the application, they choose preferences such as "Free Activities," "Outdoor Events," and a maximum budget of \$20. The system filters hundreds of available events and displays only those that match the selected criteria. A recommendation list appears alongside the map, highlighting activities that align with the user's interests.

The resident discovers a free outdoor concert and a local art walk occurring in nearby neighborhoods. After reviewing the recommendations, they save both activities and share one with a friend. The application helps the user narrow down choices and discover events they may not have found through traditional searches.

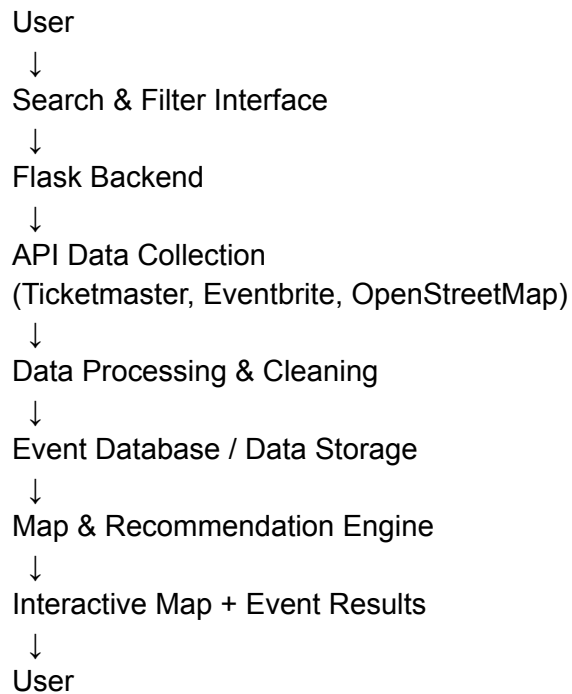
Technical Details

- Apply filtering logic based on category, cost, date, and distance.
- Store user preferences during the session.
- Generate recommendation lists from available event data.
- Update map markers dynamically as filters change.

- Remove duplicate events collected from multiple APIs.
- Optionally integrate weather data to recommend indoor or outdoor activities.

Technical flow

The flow of data would travel like this:



I also included a written data map below:

USER INTERFACE

①

- collect user data
- Inputs + display results

Input

- search → keyword (string)
- Filters →
 - Date (date object)
 - Budget (integer)
 - Category (lists)
 - Neighborhood (unknown)
- map marker
- Saving favorite events

Output

- request sent to Flask

API DATA COLLECTION

- Retrieve information from APIs

① Gets search criteria from FLASK Backend

② Processes Query API (Ticketmaster, eventbrite, OPENstreetmap)

→ **Outputs**: JSON

- Event name, dates, categories, location

* Data type = JSON dictionary, lists

DATA PROCESSING

③

- Clean + organize

NEEDS:

- Remove duplicate events
- Standardize categories
- Format dates
- addresses → coordinates (map)
- Assign neighborhood

Input: API JSON data

Output: Structured objects

- lists
- dictionaries
- Pandas

②

Map Visualization

④

- display events geographically

NEEDS:

- map markers
- neighborhood boundaries
- cluster nearby events

APIs:

- leaflet.js
- Openstreetmap

Input:

- coordinates (need to standardize)
- Neighborhood GEOJSON data

Output:

Interactive map!

(this part I am worried about but Youtube swears that leaflet will do a bulk of this)